FashionFusion

Project Team

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Contents

List of Figures

List of Tables

Chapter 1

Introduction

In this world of continuous and fast enhancement where every retailer and Brand is targeting remote customers through ecommerce. And after the pandemic of COVID-19 more sellers realised the importance of ecommerce and started selling online too. With the increasing number of shopping websites, it becomes difficult for customers to choose and shop from multiple websites. FashionFusion brings a solution for this problem by allowing customers to browse clothes of different brands at one platform, with lot of useful features for customer's ease and displays brand's image bases on their performance.

1.1 Problem Statement

When shopping online we buy clothes from different brands and while browsing clothes we have to go from one website to another to see products available with different brands and if we want to compare products, we just cannot, because most brands do not have this feature in their websites and comparing clothes from different brands becomes more difficult. Moreover most brands in Pakistan do not have review systems enabled in their platforms because they don't want to give an impression that their products are not 5 star rated. Keeping aside these problems large number of people still hesitates to buy clothes online due to unknown credibility of the stores. With FashionFusion all these Problems will be catered, it will aggregate products from different clothing brands with the feature of comparing them and allowing users to give their reviews. And for the credibility of stores this platform will also show a brand's performance score.

1.2 Scope

Products from different brands will be scraped and stored in our database. A responsive website including major functionalities of an ecommerce platform (like searching and filtering) will be built and the products stored in the database will be displayed on the website. This website will also include product comparison features and a reviews system. After creating the basic website the scraper will be set to fetch products fortnightly to keep the platform updated. Campaigns will be monitored and sentiment analysis will also be done to analyze brands' performance and create a brand's picture. For campaign monitoring prices will be regularly checked for change whether a brand provides a year long sale or it provides occasional sale and do the brands hike prices before sales. For sentiment analysis dataset will be gathered from scraping the brands' social media comments, this data will then be used to train the model. The model will then be integrated in the website. A pre-trained LLM Chatbot will then be integrated to enhance the customer's shopping experience.

1.3 Modules

The Modules of this Project are as follows:

1.3.1 Web Scraping

To start with this project first requirement is to gather products from different brands' websites and store them on a unified database. For sentiment Analysis comments and reviews fro brands' social media pages will be scraped to create a dataset.

1. Products Scraping
2. Comments and Reviews Scraping

1.3.2 Frontend Module

This is the Users' interaction interface where they can access all the functionalitis of Fashion Fusion.

1. Responsive UI / UX
2. Advanced Searching and Filters

3. Comparison Feature
4. Reviews Feature

1.3.3 Backend Module

The brains and nervous system of the FashionFusion is its secure backend which will be respondind to the user's requets sent through frontend.

1. Authentication
2. RestAPI
3. Database

1.3.4 Sentiment Analysis Module

From the scraped data (reviews and comments), data will be refined [3] and a sentiment analysis model will be trained [2] which will then be integrated in the website. This model will be kept as self training [1] so that it can update brands' score

1. Data cleaning
2. Training the Model
3. Integrating it with the Website
4. Continuous Model Training

1.3.5 LLM

User will be able to communicate with a chatbot who will give information to the user regarding a certain brand. It will also help users to easily search for products and take recommendations,

1.3.6 Report Generation Module

On the basis of campaigns/price monitoring and sentiment analysis brands scoring will be created and the brands will be ranked accordingly.

1. Brand Scoring and Ranking
2. Product Scoring

1.4 Work Division

Table 1.1: Table 1

Modules	Responsibilities		
	M Usman Ch	**M Fasih**	**M Hammad**
Web Scraping	✓	✓	
UX/UI	✓		✓
Frontend	✓		✓
Searching and Filtering		✓	
Sentiment Analysis Model Training	✓		✓
Sentiment Analysis Model Integration	✓		
Report Generation			✓
LLM Integration		✓	
Price Monitoring		✓	✓
Testing	✓	✓	✓

Chapter 2

Project Requirements

This section outlines the necessary requirements for the successful completion of the project. Project requirements can be divided into two main categories: functional requirements, which describe the system's core operations, and non-functional requirements, which specify performance, security, and usability standards.

2.1 Use-case

2.1.1 SignUp

Actors: Customer
Description: A user registers for an account on the FashionFusion platform.
Preconditions: The user has access to the internet and a valid email address.
Postconditions: A new user account is created and a confirmation email is sent.

2.1.2 Login

Actors: Customer
Description: A user logs into their FashionFusion account using their credentials.
Preconditions: The user must have a registered account.
Postconditions: The user is authenticated and granted access to the platform.

2.1.3 Search Products

Actors: Customer
Description: A user searches for products using keywords or filters.

Preconditions: The user is logged into the platform.
Postconditions: The system displays a list of products that match the search criteria.

2.1.3.1 Compare Products

Actors: Customer
Description: A user compares multiple products side by side.
Preconditions: The user has selected products for comparison.
Postconditions: The system displays a comparison of the selected products, highlighting differences and similarities.

2.1.3.2 Give Review

Actors: Customer
Description: A user submits a review for a product they have purchased.
Preconditions: The user has purchased the product and is logged into the platform.
Postconditions: The submitted review is stored in the database and associated with the product.

2.1.3.3 Review Analysis

Actors: System
Description: The system analyzes collected comments and reviews to generate sentiment scores for brands.
Preconditions: There are enough reviews and comments in the database for analysis.
Postconditions: Sentiment scores are updated and available for users to view.

2.1.3.4 Rank Brands

Actors: System
Description: The system generates scores for brands based on sentiment analysis and performance metrics.
Preconditions: The sentiment analysis model has been trained and data is available.
Postconditions: Brand scores are calculated and stored in the database for retrieval by users.

2.1.3.5 Chatbot Interaction

Actors: Customer
Description: A user interacts with the chatbot to receive information and recommendations regarding products and brands.
Preconditions: The user is logged into the platform.
Postconditions: The chatbot provides relevant information based on user queries.

2.1.3.6 Advanced Filtering

Actors: Customer
Description: A user applies filters to narrow down product search results based on specific criteria (e.g., price, category, brand).
Preconditions: The user is on the product search page.
Postconditions: The system displays filtered results based on user selections.

2.1.3.7 Manage Profile

Actors: Customer , Admin
Description: A user updates their profile information, including preferences and saved searches.
Preconditions: The user is logged into their account.
Postconditions: The user's profile is updated with new information.

2.1.3.8 Report Generation

Actors: System
Description: Report is generated based on brand scores and product performance.
Preconditions: The user has access to the report generation feature.
Postconditions: The system produces a downloadable report summarizing the requested data.

2.1.3.9 Real-Time Price Monitoring

Actors: System
Description: The system monitors product prices in real time and alerts users to changes.
Preconditions: The user has opted in for price alerts.
Postconditions: The user receives notifications about price changes.

2.1.3.10 View Activity Log

Actors: Customer
Description: A user views their activity log, including recent searches and interactions.
Preconditions: The user is logged into their account.
Postconditions: The user can see a summary of their activities on the platform.

2.2 Event Response Table

Table 2.1: Event Response Table

Event	System Response
Web Scraper initiates product scraping	The system scrapes product data from target websites and stores it in the database.
Web Scraper initiates comment and review scraping	The system gathers comments and reviews from social media and stores them in the database.
User submits registration form	The system creates a new user account and sends a confirmation email.
User submits login credentials	The system validates credentials and grants access if correct; otherwise, displays an error message.
User performs a product search	The system displays a list of products matching the search criteria.
User selects products for comparison	The system displays a comparison of the selected products, highlighting differences and similarities.
User submits a product review	The system stores the review in the database and associates it with the product.
Sentiment analysis is triggered	The system analyzes comments and reviews to generate sentiment scores for brands.
Brand scoring is triggered	The system generates scores for brands based on sentiment analysis and performance metrics.
User interacts with the chatbot	The chatbot responds with relevant information and recommendations based on user queries.
User applies filters to search results	The system displays filtered product results based on user selections.
User updates profile information	The system updates the user's profile with new information.
User requests a report	The system processes the request and produces a downloadable report summarizing brand scores and product performance.
Continuous model training is initiated	The system retrains the sentiment analysis model with new data to improve accuracy.

2.3 Functional Requirements

2.3.1 Web Scraping

Following are the requirements for the Web Scraping module:

1. The system shall scrape product information from various clothing brand websites, including details such as product name, price, description, and image URL.
2. The system shall scrape comments and reviews from the brands' social media pages to create a dataset for sentiment analysis.
3. The system shall store the scraped product data and reviews in a unified database for easy access and retrieval.

2.3.2 Frontend Module

Following are the requirements for the Frontend module:

1. The system shall provide a responsive UI/UX to enhance user interaction and accessibility across devices.
2. The system shall offer advanced searching and filtering options for users to find products easily based on various criteria (e.g., price, brand, category).
3. The system shall allow users to compare multiple products from different brands on a single screen.
4. The system shall enable users to post and view reviews for products, enhancing user feedback and interaction.

2.3.3 Backend Module

Following are the requirements for the Backend module:

1. The system shall implement user authentication to ensure secure access to the platform.
2. The system shall provide a REST API to facilitate communication between the frontend and backend components.
3. The system shall ensure a secure database to store user information, product details, and reviews.

2.3.4 Sentiment Analysis Module

Following are the requirements for the Sentiment Analysis module:

1. The system shall perform data cleaning on scraped reviews and comments to prepare them for analysis.

2. The system shall train a sentiment analysis model using the cleaned data to assess customer sentiment towards brands.

3. The system shall integrate the trained sentiment analysis model with the website to provide real-time sentiment assessment.

4. The system shall implement continuous model training to keep the sentiment analysis updated based on new data.

2.3.5 LLM (Chatbot)

Following are the requirements for the LLM module:

1. The system shall allow users to communicate with a chatbot that provides information about specific brands.

2. The system shall enable the chatbot to assist users in searching for products and providing recommendations based on user queries.

2.3.6 Report Generation Module

Following are the requirements for the Report Generation module:

1. The system shall generate brand scoring based on price monitoring and sentiment analysis results.

2. The system shall rank brands accordingly based on their scores, allowing users to see the best-performing brands.

3. The system shall generate reports detailing product scoring and brand performance metrics.

2.4 Non-Functional Requirements

2.4.1 Reliability

The following show how reliable the system would be:

1. The FashionFusion system shall maintain a Mean Time Between Failures (MTBF) of at least 30 days, ensuring that software failures occur infrequently.

2. In the event of a software failure, the system shall log the failure details and notify the technical support team within 5 minutes of detection.

3. The system shall implement error detection mechanisms to identify anomalies and provide automatic alerts for system administrators.

2.4.2 Usability

The following show the usability of the system:

1. The system shall allow users to retrieve their previously viewed products with a single interaction, enhancing the user experience.

2. The system shall provide clear and concise feedback to users for any errors encountered, helping them recover from mistakes efficiently.

3. The user interface shall be designed to require no more than three clicks to reach any main feature, ensuring an efficient navigation experience.

2.4.3 Performance

The following steps would be taken to ensure best performance:

1. The system shall ensure that 95% of product pages load completely within 3 seconds over a standard broadband connection (at least 20 Mbps).

2. The sentiment analysis module shall process and update brand scores within 10 seconds after new reviews are scraped, maintaining up-to-date information for users.

3. The system shall handle at least 100 concurrent users without performance degradation, ensuring a seamless experience during peak usage.

2.4.4 Security

This is how security will be ensured:

1. The FashionFusion system shall implement measures to protect user data and prevent unauthorized access, ensuring that only authenticated users can access sensitive information.
2. The system shall encrypt all sensitive data in transit and at rest to safeguard against potential data breaches.
3. The system shall maintain a comprehensive logging system to monitor access and changes to user accounts and sensitive data, allowing for audit trails in case of security incidents.

Chapter 3

System Overview

FashionFusion is a review-based platform designed to provide users with valuable insights into fashion brands and products. It collects data from various sources, performs sentiment analysis on user reviews, and presents the results in an interactive and user-friendly interface. The platform allows users to search and filter products based on categories, brands, and other criteria, providing a comprehensive comparison and review system. It also monitors price changes and campaigns over time, giving users an in-depth view of brand performance. With features like a chatbot for interaction, report generation, and sentiment analysis, FashionFusion aims to enhance users' understanding of the fashion market and support better decision-making.

3.1 Architectural Design

The architecture of the FashionFusion platform is based on a Layered Architecture Pattern, which divides the system into multiple layers, each responsible for specific types of functionalities. This layered approach facilitates maintainability, scalability, and modular development, ensuring that each module can be developed, tested, and updated independently.

Overview of Modules:
The platform is divided into the following primary modules:

1. User Management Module:

    - Manages user-related operations such as registration, login, and profile management.
    - Provides user authentication and authorization to ensure secure access to system functionalities.

2. Product Management Module:
    - Handles operations related to product data, such as scraping new products and updating product prices.
    - Manages the storage and retrieval of product information from the database.
3. Review Management Module:
    - Manages all review-related functionalities, including submission, editing, and deletion of reviews.
    - Stores user reviews and links them to respective products.
4. Analytics Module:
    - Processes user reviews and feedback to generate sentiment scores and track trends.
    - Analyzes product data to monitor changes in product prices and detect sales patterns.
5. Chatbot Module:
    - Provides an interactive medium for users to ask questions and receive product recommendations or information.
    - Integrates data from other modules to provide real-time responses to user queries.
6. Report Generation Module:
    - Generates detailed reports based on analytics results and product information.
    - Produces visualizations and insights for users to make informed decisions.

Module Interactions and Relationships The modules interact with each other as follows:

- The User Management Module provides authenticated access to the platform and collaborates with the Product Management and Review Management modules to store user activities, such as product views and review submissions.
- The Product Management Module and Review Management Module serve as data sources for the Analytics Module, which processes this data to generate insights such as sentiment scores and trends.
- The Analytics Module communicates with the Report Generation Module to produce comprehensive reports, providing users with an easy-to-understand summary of findings.

- The Chatbot Module pulls data from multiple modules to deliver accurate responses to user queries, ensuring seamless interaction.

3.1.1 Architectural Style and Module Mapping

The FashionFusion platform follows a Layered Architecture Pattern consisting of the following layers:

1. **Presentation Layer:**
   - **Modules:** User Management Module, Chatbot Module, Report Generation Module.
   - **Purpose:** Handles all user interactions and presents data to the user.
2. **Business Logic Layer:**
   - **Modules:** Product Management Module, Review Management Module, Analytics Module.
   - **Purpose:** Contains the core functionality and business rules of the platform.
3. **Data Access Layer:**
   - **Modules:** Product Database, User Reviews Database.
   - **Purpose:** Manages storage and retrieval of data, supporting the business logic.
4. **External Interface Layer:**
   - **Module:** Product Scraper.
   - **Purpose:** Interacts with external sources, such as brand websites, to scrape product data.

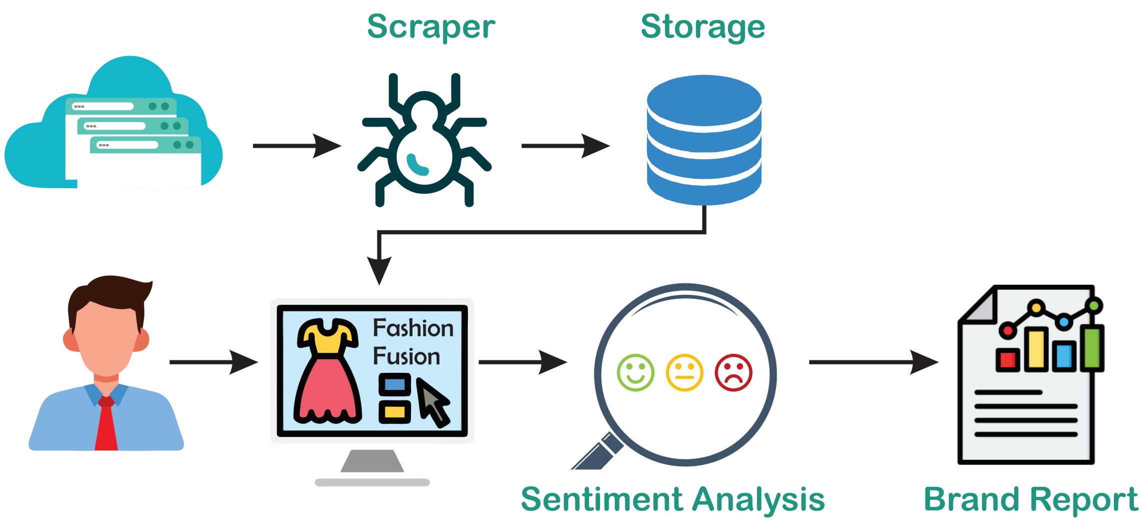

Figure 3.1: Architectural Design Diagram

3.2 Data Design

In this section, we will explain how the information domain of the FashionFusion platform is transformed into data structures, detailing how the major data entities are stored, processed, and organized.

Information Domain Transformation
The information domain of the FashionFusion system consists of various entities that represent users, products, reviews, and system-generated insights. These entities are organized into structured data formats, such as tables in a relational database, to facilitate efficient storage, retrieval, and processing.

Major Data Entities

1. User Entity:
   - Attributes: User ID, Name, Email, Password, Profile Information, etc.
   - Storage: Stored in the Users table of the database.
2. Product Entity:
   - Attributes: Product ID, Name, Description, Price, Availability, Brand, etc.
   - Storage: Stored in the Products table, updated regularly through the scraping processes.

3. Review Entity:

    - Attributes: Review ID, User ID, Product ID, Rating (1-5), Review Text, Timestamp, etc.
    - Storage: Stored in the Reviews table, allowing users to submit and manage their reviews.

4. Analytics Data:

    - Attributes: Metric ID, Product ID, Sentiment Score, Review Count, etc.
    - Storage: Generated and stored in an Analytics table for reporting purposes.

5. Chatbot Interaction Logs:

    - Attributes: Log ID, User ID, Query, Response, Timestamp, etc.
    - Storage: Stored in a Chatbot Logs table to analyze user interactions and improve the recommendation system.

Data Storage
The FashionFusion platform utilizes a relational database management system (RDBMS) for data storage. The major data storage items include:

- **User Database:** Holds user credentials and profile information.
- **Product Database:** Contains detailed information about products scraped from various websites.
- **Review Database:** Stores user-generated reviews linked to specific products.
- **Analytics Database:** Stores computed metrics and insights from the reviews and product data.

3.3 Domain Model

The domain model for **FashionFusion** encapsulates the key entities, their attributes, and the relationships that define the interactions within the platform. This model provides a clear blueprint for understanding how different components of the system interact and supports the overall functionality of **FashionFusion**.

3.3.1 Entities and Attributes

1. **User**
    - **Attributes:**
        - `UserID` (Primary Key): Unique identifier for each user.
        - `Name`: Full name of the user.
        - `Email`: User's email address.
        - `Password`: Encrypted password for authentication.
        - `ProfilePicture`: URL or path to the user's profile image.
        - `RegistrationDate`: Date when the user registered.
    - **Description:** Represents individuals who interact with the FashionFusion platform, including submitting reviews and engaging with the chatbot.
2. **Product**
    - **Attributes:**
        - `ProductID` (Primary Key): Unique identifier for each product.
        - `Name`: Name of the product.
        - `Description`: Detailed description of the product.
        - `Brand`: Brand associated with the product.
        - `Price`: Current price of the product.
        - `Availability`: Stock status (e.g., In Stock, Out of Stock).
        - `Category`: Category under which the product falls (e.g., Shoes, Clothing).
        - `ImageURL`: URL or path to the product image.
    - **Description:** Represents fashion products available on the platform, including details necessary for user reviews and analytics.
3. **Review**
    - **Attributes:**
        - `ReviewID` (Primary Key): Unique identifier for each review.
        - `UserID` (Foreign Key): Identifier linking to the user who submitted the review.
        - `ProductID` (Foreign Key): Identifier linking to the reviewed product.
        - `Rating`: Numerical rating (e.g., 1 to 5 stars).
        - `ReviewText`: Textual content of the review.
        - `Timestamp`: Date and time when the review was submitted.

- **Description:** Captures user-generated feedback on products, essential for sentiment analysis and analytics.

4. **Analytics**

- **Attributes:**
  - `MetricID` (Primary Key): Unique identifier for each analytics record.
  - `ProductID` (Foreign Key): Identifier linking to the associated product.
  - `SentimentScore`: Calculated sentiment score based on reviews.
  - `ReviewCount`: Total number of reviews for the product.
  - `PriceTrend`: Historical price data over time.
- **Description:** Stores processed data derived from user reviews and product information to provide actionable insights.

5. **ChatbotInteractionLog**

- **Attributes:**
  - `LogID` (Primary Key): Unique identifier for each chatbot interaction.
  - `UserID` (Foreign Key): Identifier linking to the user interacting with the chatbot.
  - `Query`: The user's input or question.
  - `Response`: The chatbot's reply.
  - `Timestamp`: Date and time of the interaction.
- **Description:** Logs interactions between users and the chatbot to improve response accuracy and user experience.

3.3.2 Relationships

1. **User to Review:**

- **Type:** One-to-Many.
- **Description:** A single user can submit multiple reviews, but each review is associated with only one user.

2. **Product to Review:**

- **Type:** One-to-Many.
- **Description:** A single product can have multiple reviews, but each review pertains to only one product.

3. **Product to Analytics:**

    - **Type:** One-to-One.
    - **Description:** Each product has one corresponding analytics record that aggregates review data and trends.

4. **User to ChatbotInteractionLog:**

    - **Type:** One-to-Many.
    - **Description:** A single user can have multiple chatbot interaction logs, but each log entry is linked to one user.

3.4 Design Models

This section outlines the design models applicable to the current iteration of the **Fashion-Fusion** project. The models provide a visual and structured representation of the system's behavior, structure, and interactions.

3.4.1 Activity Diagram

The activity diagram represents the workflow for submitting a product review by a user. It highlights the sequence of actions and decisions involved in the process. The detailed diagram is provided in Appendix D.1.

3.4.2 Class Diagram

The class diagram outlines the system's static structure, showing the entities (classes) identified in Section **??** along with their attributes, methods, and relationships. The complete diagram can be found in Appendix D.2.

3.4.3 Class-level Sequence Diagram

The sequence diagram focuses on the interaction between objects for the *Review Submission* use case. It illustrates the exchange of messages to complete the process. The full diagram is included in Appendix D.3.

3.4.4 State Transition Diagram

The state transition diagram models the different states of a product review and the events that trigger state transitions. A detailed version of this diagram is available in Appendix D.4.

3.4.5 Design Model Summary

The diagrams presented provide a comprehensive understanding of the **FashionFusion** system up to the current iteration. These models ensure clarity in the design process and serve as a reference for further development and testing.

Chapter 4

Implementation and Testing [Up to the Current Iteration Only]

This chapter provides an overview of the design, functionality, and testing of the Fashion Fusion system. It outlines the algorithms implemented, external APIs/SDKs utilized, and the testing methodologies applied.

4.1 Algorithm Design

Fashion Fusion employs algorithms to ensure efficient functionality for its core modules. Below is a pseudocode representation of the primary algorithm used for the recommendation module.

Example: Recommendation Algorithm [H] Fashion Recommendation Algorithm [1] User preferences, available product data Generate personalized recommendations **Input:** User's selected preferences P, Product database D Filter products $F \leftarrow \text{Filter}(D, P)$ Rank products $R \leftarrow \text{Rank}(F)$ based on relevance Output top n recommendations $O \leftarrow R[1:n]$

`images/AlgorithmExample.png`

Figure 4.1: Example of Algorithm Design for Fashion Fusion

4.2 External APIs/SDKs

Fashion Fusion integrates various third-party APIs/SDKs for enhanced functionality. Table 4.1 provides a detailed list of these APIs.

API and Version	**Description**	**Purpose of Usage**	**API point/Function/C Used**
OpenAI API (2024-01-01)	NLP-based fashion analysis	Product descriptions and trends analysis	`openai.Complet`
Cloudinary	Image and video management	Product image uploads	`https://api.cl`
FastAPI	API development framework	Backend API implementation	`fastapi.APIRou` `dependencies`

Table 4.1: APIs/SDKs Used in Fashion Fusion

4.3 Testing Details

Testing ensures the Fashion Fusion system functions as intended. This section details the types of testing performed.

4.3.1 Unit Testing

Unit testing involves validating individual components of the Fashion Fusion system, such as API endpoints and algorithms.

images/UnitTestingExample.png

Figure 4.2: Example of Unit Testing

4.3.2 Integration Testing

Integration testing checks the interaction between Fashion Fusion's modules, such as recommendation systems and APIs.

4.3.3 Performance Testing

This evaluates the system's responsiveness and load handling, ensuring seamless user experiences.

4.4 Summary

This chapter summarized the design, implementation, and testing of the Fashion Fusion project, emphasizing the algorithms, external APIs, and testing methodologies utilized.

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Appendix A

Appendices

A.1 Appendix A

A.1.1 Use Case Diagram

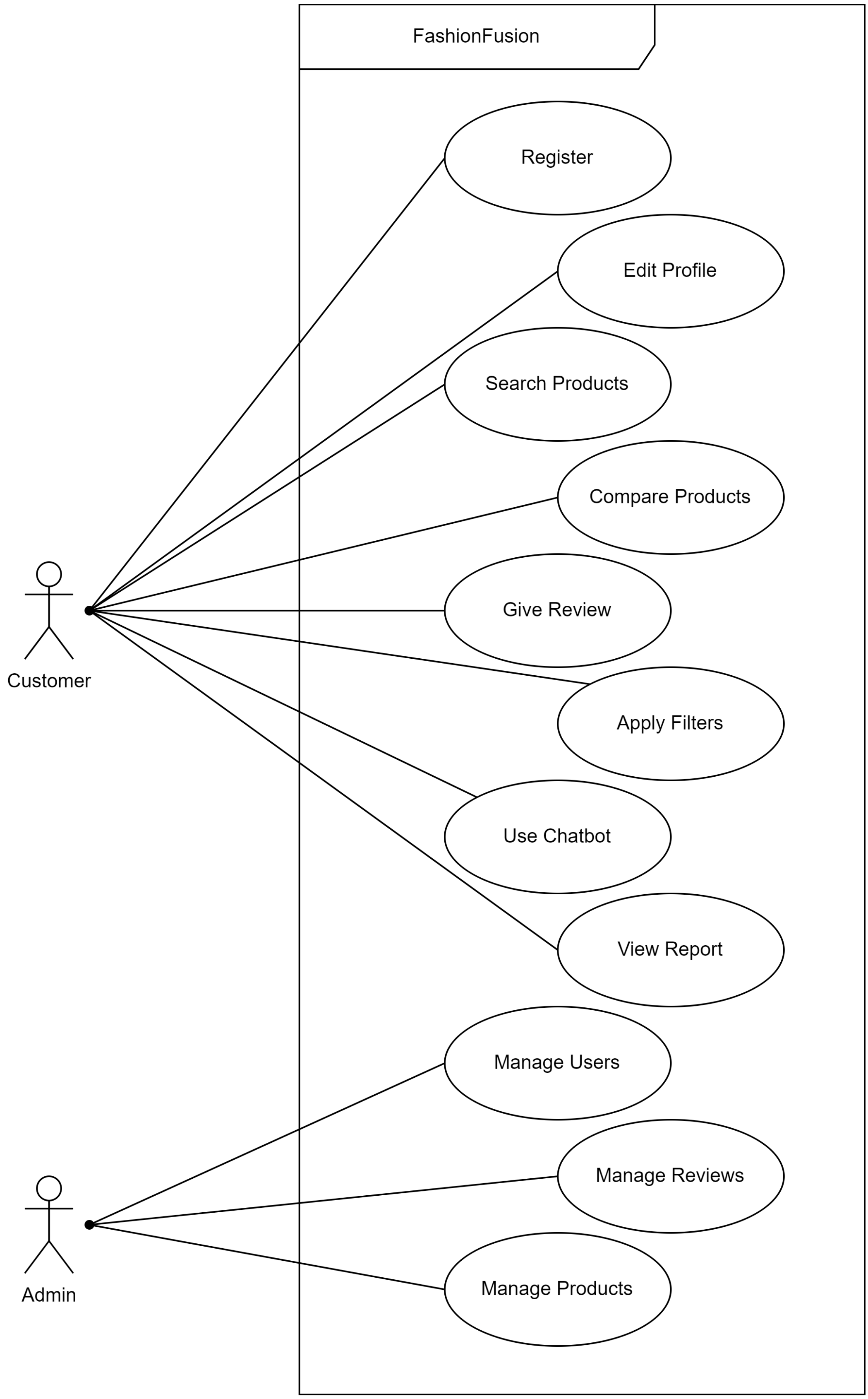

Figure A.1: UseCase Diagram

A.1.2 Event-Response Table for a Highway Intersection

Table A.1: Event Response Table

Event	System Response
Web Scraper initiates product scraping	The system scrapes product data from target websites and stores it in the database.
Web Scraper initiates comment and review scraping	The system gathers comments and reviews from social media and stores them in the database.
User submits registration form	The system creates a new user account and sends a confirmation email.
User submits login credentials	The system validates credentials and grants access if correct; otherwise, displays an error message.
User performs a product search	The system displays a list of products matching the search criteria.
User selects products for comparison	The system displays a comparison of the selected products, highlighting differences and similarities.
User submits a product review	The system stores the review in the database and associates it with the product.
Sentiment analysis is triggered	The system analyzes comments and reviews to generate sentiment scores for brands.
Brand scoring is triggered	The system generates scores for brands based on sentiment analysis and performance metrics.
User interacts with the chatbot	The chatbot responds with relevant information and recommendations based on user queries.
User applies filters to search results	The system displays filtered product results based on user selections.
User updates profile information	The system updates the user's profile with new information.
User requests a report	The system processes the request and produces a downloadable report summarizing brand scores and product performance.
Continuous model training is initiated	The system retrains the sentiment analysis model with new data to improve accuracy.

A.2 Appendix B

A.2.1 Domain Model

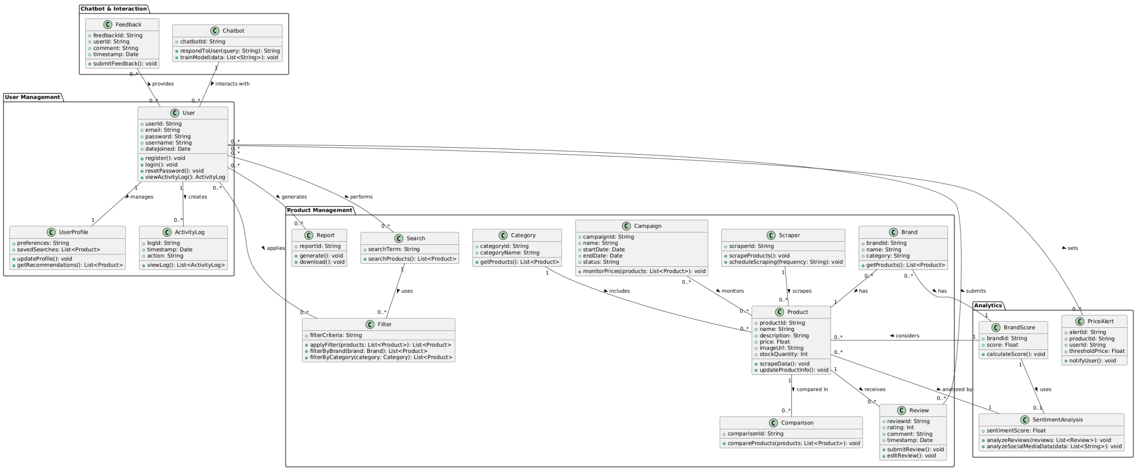

Figure A.2: Domain Model

A.3 Appendix C

A.3.1 Box And Line

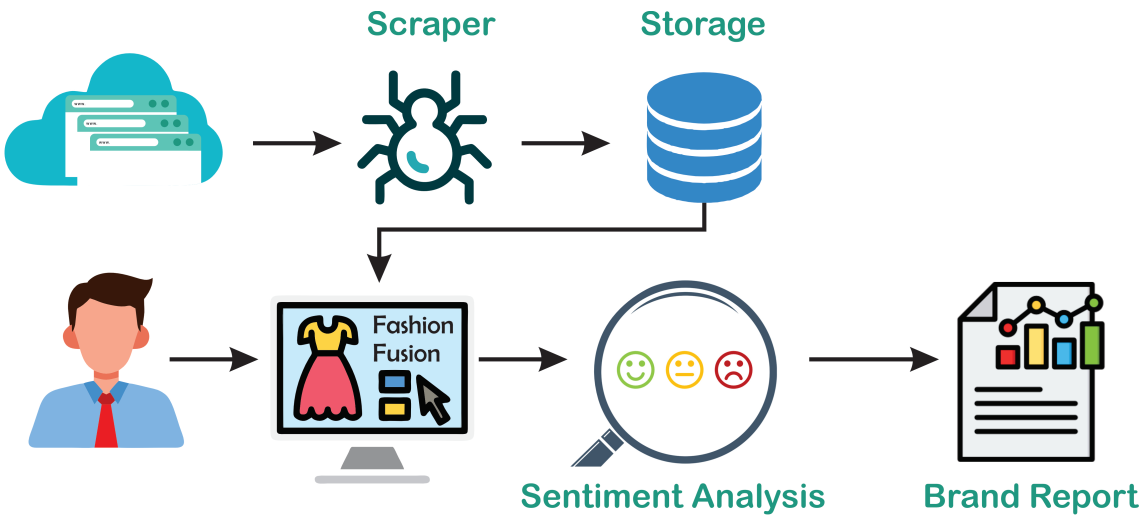

Figure A.3: Architectural Design Diagram

A.4 Appendix D

A.4.1 Activity Diagram

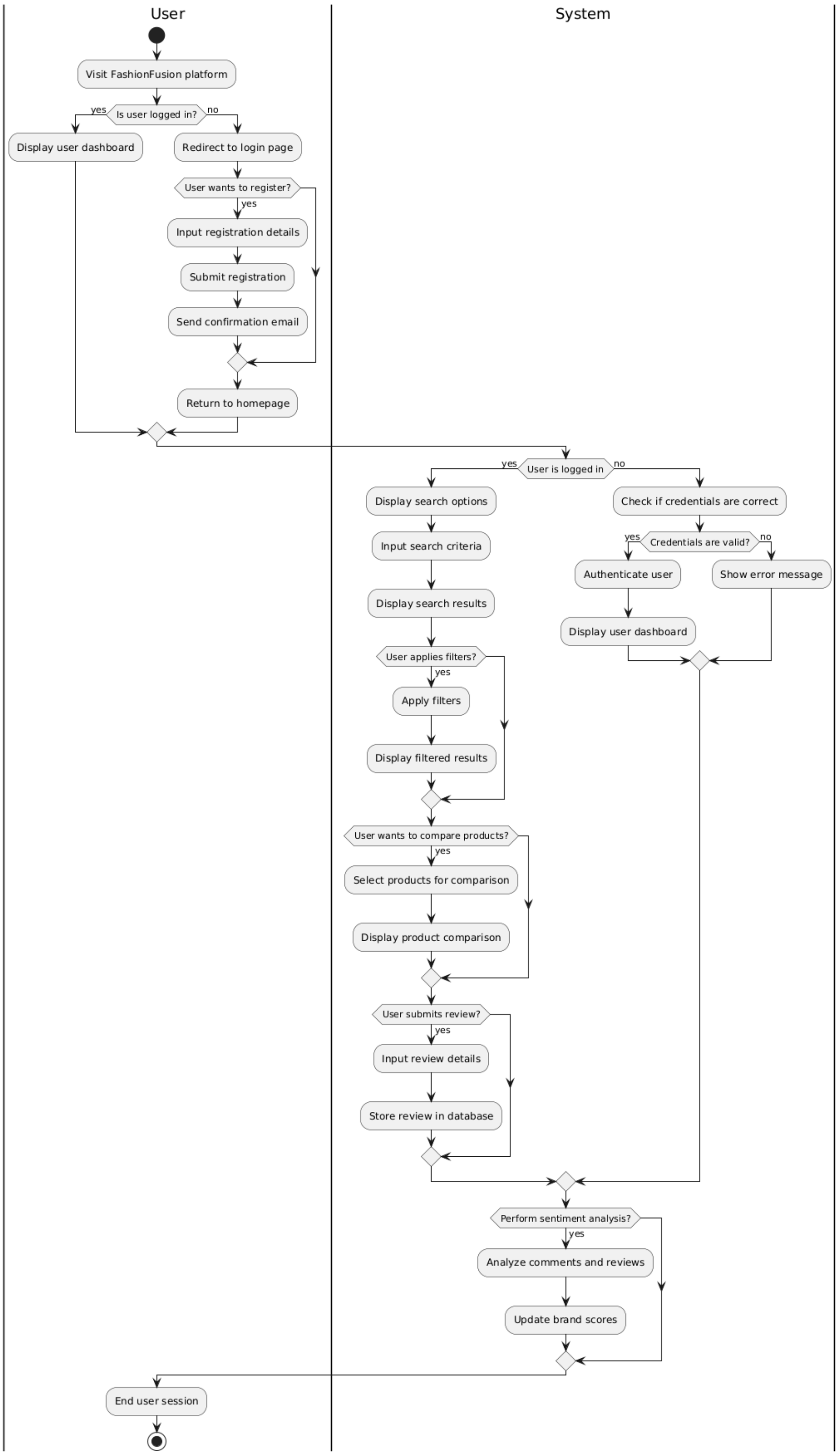

Figure A.4: Activity Diagram

A.4.2 Sequence Diagram

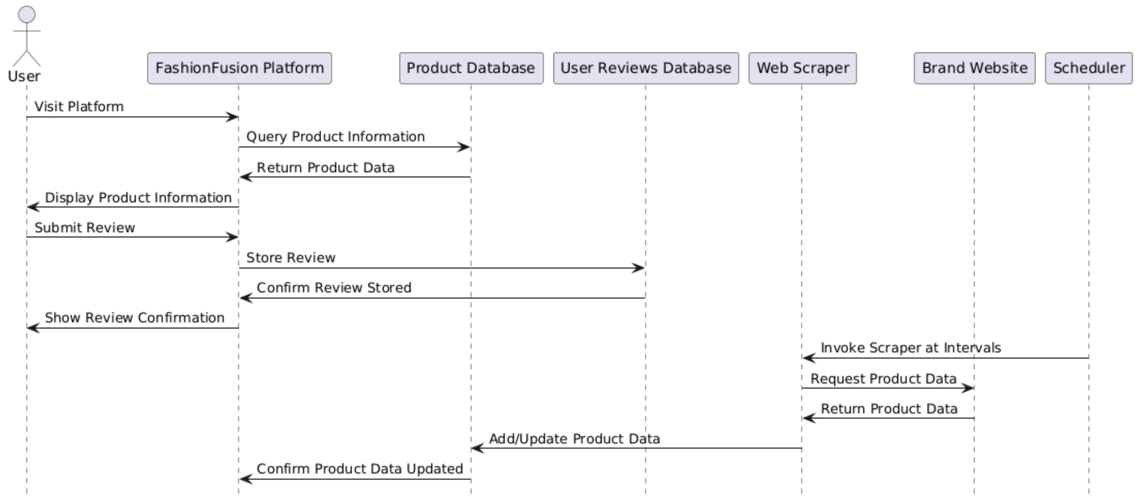

Figure A.5: System Sequence Diagram

A.4.3 Data Flow Diagram

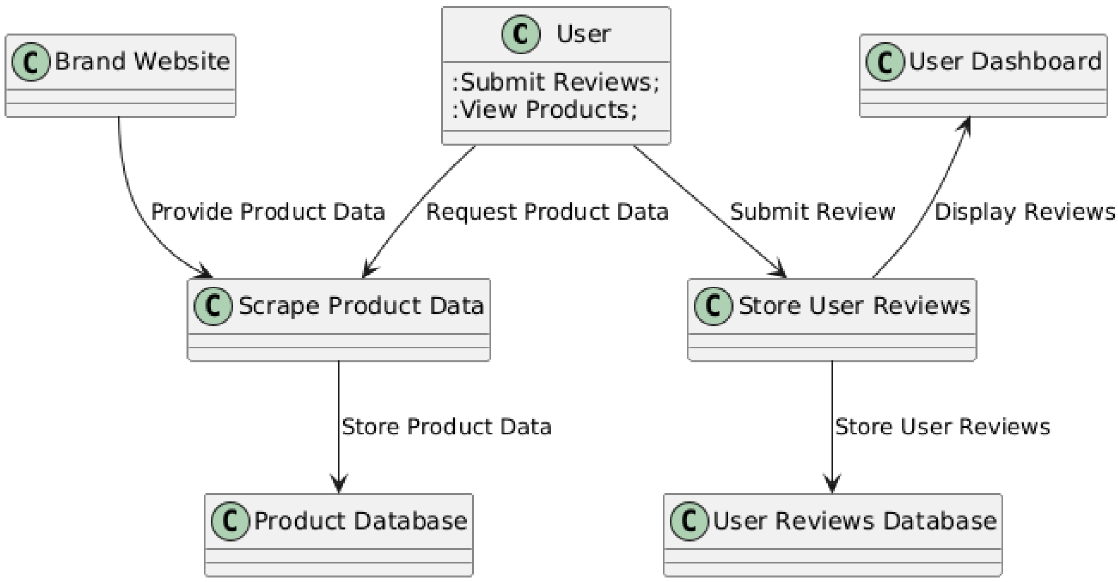

Figure A.6: Level 0 Data Flow Diagram

A.4.3.1 Components

Processes:

- **Process 1: Scrape Product Data**
  Description: This process involves collecting product information from various brand websites and storing it in the database.

- **Process 2: Store User Reviews**
  Description: This process captures user feedback and stores it in the user reviews database.

Sources/Sinks:

- **Source: Brand Website**
  Description: The external websites from which product data is scraped.

- **Sink: User Dashboard**
  Description: The interface where users can view products, reviews, and their profile information.

Data Stores:

- **Data Store 1: Product Database**
  Description: A repository for all product information scraped from various websites.

- **Data Store 2: User Reviews Database**
  Description: A storage location for user-generated reviews and feedback.

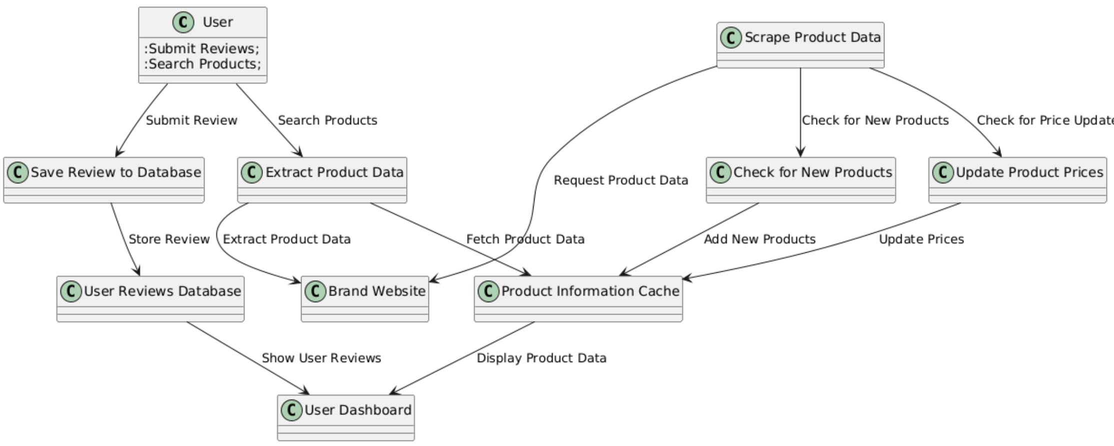

Figure A.7: Level 1 Data Flow Diagram

A.4.3.2 Components

Processes

- **Process 1: Scrape Product Data**
  Description: This process encompasses activities related to retrieving product data from brand websites, including checking for new products and updating existing product prices.

- **Process 2: Check for New Products**
  Description: This subprocess checks the brand websites for any newly added products and stores them in the product information cache.

- **Process 3: Update Product Prices**
  Description: This subprocess verifies whether the prices of existing products have changed and updates the product information cache accordingly.

- **Process 4: Extract Product Data**
  Description: This process extracts detailed product information based on user search queries and displays it in the user dashboard.

- **Process 5: Save Review to Database**
  Description: This process captures user-submitted reviews and stores them in the User Reviews Database.

Sources/Sinks

- **Source: Brand Website**
  Description: The external websites from which product data is scraped.

- **Source: User**
  Description: Individuals interacting with the system to submit reviews or search for products.

- **Sink: User Dashboard**
  Description: The interface where users can view their submitted reviews and search results.

Data Stores

- **Data Store 1: Product Information Cache**
  Description: Temporarily holds scraped product data before it is stored permanently in the main product database.

- **Data Store 2: User Reviews Database**
  Description: Stores user-generated reviews and feedback, allowing for easy retrieval and display in the user dashboard.